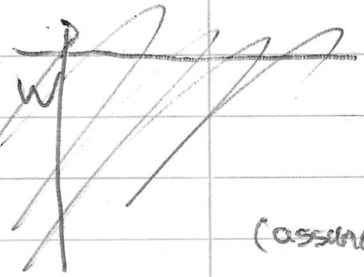
Goal → Predict Performance of RF die ~~of a~~ over a lumped model.
↳ extract ~~var~~ s2p values from ~~VV~~ VNA measurements compare →

- Ref – Microwave engineering → 4th ed

↳ Known values → reference: Woulter BBEP + CPWG Klayout Design.

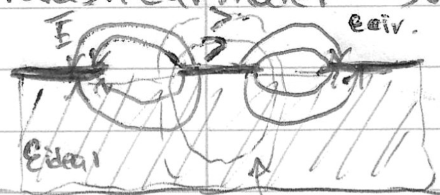


W	80μm
S	~~20~~ 25μm
L	1mm
H	1mm.
(assume) glass → ε_r	7.75
(thickness) t_{Au}	100nm.
t_{Ti}	50nm
ρ_{Au}	$3.64 \cdot 10^{-8}\,\Omega m$
ρ_{Ti}	$3.99 \times 10^{-8}\,\Omega m$.

parasitics

- R_1: conductor (Au/Ti) series resistance of 1mm trace.
- L_1: External + internal inductance of the 1mm CPWG.
- C_3: lateral (signal ↔ coplanar grounds) capacitance – main CPWG shunt C.
- C_1: vertical capacitance through glass under strip. ("extra term")
- R_2: leakage through glass substrate.
- C_2: slow-wave/space charge capacitance in lossy Si

Parasitical model → simulation (something coplanar line calculator doesn't include.
↳ not present in glass



R1 DC resistance for each layer.

$R_i = \rho_i \frac{L}{W t_i}$ → as one on top of other Au/Ti

→ parallel → $R_1 = \left(\frac{1}{R_{Au}} + \frac{1}{R_{Ti}}\right)^{-1}$.

$R_1 = 3.12\,\Omega$

L_1 external + internal inductance of the CPWG:

- external: magnetic field surrounding the current path / air/glass region. (W, S, H) geometry and (permeability $\mu(\mu_0)$) impact.
- internal: magnetic field → penetrates in metal (finite conductivity of metal).
  → it causes additional inductance at (lower frequency).
  ↳ (vanishes as skin effect → pushes it to surface.
  ↳ not very relevant

- modulus k: $k = \frac{W}{W+2S} = \frac{80\mu}{80\mu + 2(25\mu)} = 0.615\ldots$ $k' = \sqrt{1-k^2} = 0.788\ldots$ geometric factor.
- elliptical integral: $K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1-k^2\sin^2\theta}}$ → $K(k) = 1.659$; $K(k') = 1.841$
- quasi-static Z impedance: $Z_0 = \frac{30\pi}{\sqrt{\varepsilon_{eff}}} \frac{K(k')}{K(k)}$ (low frequency impedance). (behaves as resistive network → low capacitive effects).
- Phase velocity: $v = \frac{c}{\sqrt{\varepsilon_{eff}}}$ → $\varepsilon_{eff} = \frac{\varepsilon_r + 1}{2} = \frac{7.75+1}{2} = 4.375$.
- per unit length inductance: $L' = \frac{Z_0}{v}$
- lumped value: $L_1 = L' \cdot L$

$L = 1mm$; $L' = 3.52 \cdot 10^{-7}\,Hm^{-1}$

→ $L_1 = 0.352nH$

~~loaderal strip~~ above signal capacitance.

C_3 : ~~lateral~~ $C' = \frac{1}{Z_0 v}$ velocity $C_3 = C' \cdot L \rightarrow C_3 = 0.138pF$

C_1 : ~~through~~ ~~loss~~ vertical capacitance under strip (oxide).

$$C_1 = \varepsilon_0 \varepsilon_r \frac{(W+2S)L}{H}$$

$\varepsilon_0 = 8.854 \cdot 10^{-12}$

$\dots \rightarrow C_1 = 8.92 fF$

R_2 : glass is a pretty good substrate compared to Si in such case.
leakage (modeled as resistance) through glass.

$$R_2 = \rho_{glass} \cdot \frac{H}{(W+2S)L}$$

(borosilicate glass) commonly used for interposer

glass $\rho = 10^{12} - 10^{14}\ \Omega\,cm$
→ effectively "open"

→ so choosing $1T\Omega$

C_2 : slow wave/space charge → irrelevant to glass so.
$0pF$.